Normal-Model z-Scores and Percentiles
Answer Key

Precalculus / Advanced Statistics

Quick Answers

1. 1.4
2. -0.7
3. 0.8413
4. 0.9332, 0.0668
5. 1.5, 0.9332
6. 0.1587, 0.7745
7. 0.8997, 180.24
8. 0.9772, 700
9. 1.5, —
10. $\frac{57}{2500}$
11. 18, 4, 1.5
12. -2, 0.0228, 0.1003, 498.2

Curriculum

Level: Precalculus / Advanced Statistics
HSS-IC.A–HSS-IC.B – *problems 10, 12*
HSS-ID.A–HSS-ID.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 11*
Difficulty 1–5 of 5 across 25 tagged checks.

1. National test, $\mu = 500$, $\sigma = 100$. Find the z-score of 640 and interpret it.

Solution. Standardising, $z = \dfrac{640 - 500}{100} = \dfrac{140}{100}$. The score sits 1.4 standard deviations ABOVE the mean. That single number is what makes scores from different scales comparable — it says nothing about the raw units.

$$\boxed{z = 1.4}$$

2. Same test. Find the z-score of 430.

Solution. $z = \dfrac{430 - 500}{100} = \dfrac{-70}{100}$, so the score is 0.7 standard deviations BELOW the mean. The sign carries the direction; a student who computes $\dfrac{\mu - x}{\sigma}$ gets $+0.7$ and reports the score as above average, which is the misconception this problem is built to catch.

$$\boxed{z = -0.7}$$

3. What proportion of a normal distribution lies below $z = 1.00$, and what percentile is that?

Solution. Read the column headed 1.00: the area to the left is 0.8413. Since 84.13% of the distribution lies below that point, the value sits at roughly the 84th percentile. "Percentile" is the same information as "area to the left", expressed as a percentage.

$$\boxed{0.8413}$$

4. What proportion lies below $z = 1.50$, and what proportion lies above it?

Solution. The table gives the area to the LEFT of 1.50 as 0.9332. The whole distribution has area 1, so the area to the right is $1 - 0.9332 = 0.0668$. Reporting 0.9332 as the answer to the second half is the standard error: the table is always a left-tail area, never a right-tail one.

$$\boxed{0.9332 \text{ below},\ 0.0668 \text{ above}}$$

5. Heights $\mu = 170$ cm, $\sigma = 8$ cm. Find z for 182 cm and the proportion shorter than that.

Solution. $z = \dfrac{182 - 170}{8} = \dfrac{12}{8} = 1.5$. Because the table gives left-tail areas and "shorter than" is a left-tail question, read the entry directly: the proportion of adults shorter than 182 cm is 0.9332, about 93%.

$$\boxed{z = 1.5,\ 0.9332}$$

6. Proportion of heights between 162 cm and 182 cm.

Solution. Standardise both endpoints: $z_1 = \dfrac{162 - 170}{8} = -1.00$ and $z_2 = \dfrac{182 - 170}{8} = 1.50$. The area BETWEEN two values is the difference of their left-tail areas, larger minus smaller: $0.9332 - 0.1587$. So about 77.45% of adults fall in that interval. Adding the two entries instead of subtracting is the usual slip — a sum bigger than 1 is the tell.

$$\boxed{0.9332 - 0.1587 = 0.7745}$$

7. Which height is at the 90th percentile? ($\mu = 170$ cm, $\sigma = 8$ cm.)

Solution. The 90th percentile is the value with area 0.90 to its left. The closest table entry is 0.8997, at $z = 1.28$. Now reverse the standardising with $x = \mu + z\sigma$:

$$x = 170 + 1.28(8) = 170 + 10.24.$$

The 90th-percentile height is about 180.24 cm. Adding 1.28 straight onto the mean — forgetting to multiply by σ — gives 171.28 cm and is the trap built into this problem.

$$\boxed{0.8997 \to z = 1.28,\ x = 180.24}$$

8. Scholarship for the top 2.28% on the national test. Find the cut-off score.

Solution. Top 2.28% means area 0.0228 to the RIGHT, so area $1 - 0.0228 = 0.9772$ to the left. The table gives that area at $z = 2.00$. Then

$$x = 500 + 2(100) = 700.$$

A candidate needs at least 700. Note the first move: convert the right-tail statement into a left-tail area, because that is the only thing the table reports.

$$\boxed{0.9772 \to z = 2,\ x = 700}$$

9. Ana: 640 on the test ($\mu = 500$, $\sigma = 100$) and 28 on an exam with $\mu = 22$, $\sigma = 4$. Which is relatively stronger?

Solution. First result: $z = \dfrac{640 - 500}{100} = 1.4$. Second result: $z = \dfrac{28 - 22}{4} = \dfrac{6}{4} = 1.5$. Comparing, $1.4 < 1.5$, so the SECOND result is the relatively stronger one: it sits further above its own mean measured in that distribution's own standard deviations. Comparing the raw scores 640 and 28 is meaningless — they are on different scales, which is exactly why standardising exists.

$$\boxed{1.4 < 1.5}$$

10. About 228 of every 10,000 candidates score above 700. Probability a random candidate did?

Solution. The probability is the relative frequency $\frac{228}{10000}$. Dividing numerator and denominator by 4 gives $\frac{57}{2500}$, which is 0.0228 — the same right-tail area found in problem 8, as it must be, since 700 is the $z = 2.00$ cut-off.

$$\boxed{\tfrac{57}{2500} = 0.0228}$$

11. Squad scores 12, 14, 16, 18, 20, 22, 24 (whole population). Find μ, σ, and the z-score of 24.

Solution. (a) The seven values are evenly spaced, so the mean is the middle one: $\frac{12+14+16+18+20+22+24}{7} = \frac{126}{7} = 18$.

(b) Deviations from 18 are $-6, -4, -2, 0, 2, 4, 6$; their squares sum to $36 + 16 + 4 + 0 + 4 + 16 + 36 = 112$. Dividing by $n = 7$ (population, because the squad IS the population) gives a variance of 16, so $\sigma = 4$.

(c) $z = \frac{24 - 18}{4} = \frac{6}{4} = 1.5$. Dividing by $n-1$ instead would give $\sigma \approx 4.32$ and a different z; the problem says to treat the squad as the whole population, so n is correct here.

$$\boxed{\mu = 18,\ \ \sigma = 4,\ \ z = 1.5}$$

12. Bottles: $\mu = 500$, $\sigma = 2.5$, label 495. (a) z; (b) proportion underfilled; (c) new mean; (d) the assumption.

Solution. (a) $z = \frac{495 - 500}{2.5} = \frac{-5}{2.5} = -2$.

(b) "Underfilled" is a left-tail question, so read the table at $z = -2.00$: about 0.0228 of bottles, a little over two in a hundred.

(c) An accepted proportion of 0.1003 to the left corresponds to $z = -1.28$. We need 495 to sit at that z under the new mean, so from $z = \frac{x-\mu}{\sigma}$ we get $\mu = x - z\sigma = 495 - (-1.28)(2.5) = 495 + 3.2 = 498.2$. Note this LOWERS the mean, allowing more underfilled bottles, not fewer — 0.1003 is a looser standard than the 0.0228 found in (b), which is worth pointing out to a student who expects every adjustment to tighten quality.

(d) Everything above assumes the fill volumes really follow a normal model. If the distribution were skewed — say a jammed valve produced a cluster of very low fills — the table areas would not describe it, and every proportion quoted here would be wrong even though the arithmetic was right.

$$\boxed{z = -2,\ \ 0.0228;\ \ 0.1003 \to \mu = 498.2}$$